ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

MECHATRONICS

PAPER 1 · PRACTICE C

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 4. Laid out like the DHET Mechatronic Systems paper (project based, 100 marks): feasibility and specifications, an automatic car-park boom gate, torque and gearbox calculations, forward/reverse motor control with interlocks and limit switches, vehicle sensors, flowcharts, budgets, drawings, system block diagrams, commissioning, snag lists, fault-finding and maintenance.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FIVE questions. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show ALL formulae, substitutions and calculations, and write units with all answers. Take g = 9,81 m/s².
5. Draw sketches, charts and circuits neatly. Use only a black or blue pen.

> **PROJECT.** A shopping centre wants an automatic boom gate at the entrance of its staff car park. When a car stops on the detector loop and the driver presses the button (or swipes a card), a motor and gearbox lift the boom. The boom stays open until the car has passed the photo-electric safety beam and then closes again. Limit switches stop the motor at the fully open and fully closed positions, and the boom must never come down on a car.

QUESTION 1

1.1 Explain the purpose of a feasibility study before a project starts. (2)

1.2 Name FOUR headings (aspects) of the design specification for the boom gate, and write ONE requirement under each heading. (8)

[10]

QUESTION 2

2.1 Make a neat freehand sketch of the boom gate. Show the boom and its pivot, the housing with the motor and gearbox, the limit switches, the detector loop and the photo-electric beam. (5)

2.2 The boom has a mass of 12 kg, and its centre of gravity is 2 m from the pivot. It is lifted through a gearbox with a ratio of 50 : 1 and an efficiency of 80%.

2.2.1 Calculate the torque needed at the pivot to lift the boom from the horizontal position. (3)

2.2.2 Calculate the torque the motor must deliver. (3)

2.2.3 The boom must open through 90° in 3 seconds. Calculate the average motor speed in r/min. (2)

2.3 FIGURE 1 shows the existing circuit, which can only lift the boom. Redraw and modify it so that the motor can also lower the boom: contactor K1 lifts (UP) and contactor K2 lowers (DOWN). Use electrical interlocking, and add limit switch S3 (fully open) and limit switch S4 (fully closed). (8)

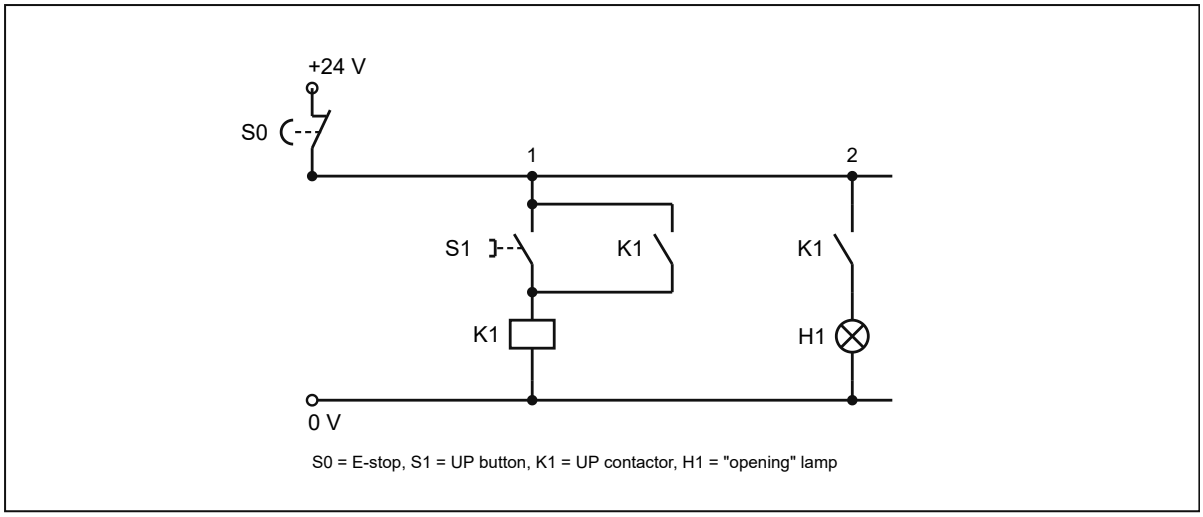


FIGURE 1: Existing boom motor circuit (UP only)

2.4 Explain why the boom must stop and open again if the safety beam is broken while it is closing. (2)

2.5 Name TWO types of sensor that can detect a vehicle at the gate. (2)

[25]

QUESTION 3

3.1 Draw a flowchart for the operation of the boom gate, from a car arriving to the boom being closed again. Include the safety beam. (10)

3.2 Draw up a budget for the boom gate with suitable column headings and at least SIX items. (8)

3.3 Explain the difference between a part drawing and an assembly drawing, and when EACH is used. (4)

3.4 Draw a system block diagram of the boom gate, showing its inputs, controller and outputs. (6)

3.5 Explain why the project must have a clear time plan (schedule). (2)

[30]

QUESTION 4

4.1 Use a spider diagram to show SIX areas that must be tested on the boom gate. (6)

4.2 Explain how the limit switches are set during commissioning. (3)

4.3 Explain what a snag list is. (2)

4.4 List FOUR things in which the security staff must be trained when the gate is handed over. (4)

[15]

QUESTION 5

5.1 Draw up a maintenance schedule for the boom gate with FOUR tasks and how often each must be done. (8)

5.2 The boom opens but does not close. Give FOUR possible causes, and explain how you would check EACH. (8)

5.3 Explain why the oil in the gearbox must be checked regularly. (2)

5.4 Give TWO advantages of remote monitoring of the boom gate. (2)

[20]

TOTAL: 100

END OF QUESTION PAPER